
INTEGRATION PROPOSAL

Sendaza Liquidity Engine

Multi-asset purchase pricing, MPC-secured treasury custody, and external blockchain withdrawals for Sendaza.

Prepared for: Sendaza

Initial environment: Sandbox with configuration-driven assets, fiat currencies, and networks

Custody: Fireblocks Developer Sandbox using MPC

Status: Design approval and integration planning

Date: 29 August 2026

1. Executive Summary

Sendaza manages what customers own. SLE prices, reserves, safeguards, and delivers the crypto that backs that ownership.

The Sendaza Liquidity Engine (SLE) is a private service that enables Sendaza customers to purchase configured crypto assets using configured fiat balances and withdraw supported assets over approved blockchain networks.

Sendaza remains the customer and accounting system of record. It owns authentication, KYC, transaction authorization, balance locking, and double-entry ledger entries. SLE owns market-data conversion, executable quotes, spreads, purchase fees, treasury availability, inventory reservations, MPC custody integration, blockchain execution, confirmation monitoring, and treasury reconciliation.

The first deliverable serves Sendaza only and runs with test balances, testnet assets, and a Fireblocks Developer Sandbox. External client onboarding and public partner APIs are deferred. The sandbox is not approved for mainnet or real customer funds.

2. Business Deliverable

The sandbox MVP will demonstrate one complete configured crypto/fiat journey:

- Request a short-lived executable quote for an enabled crypto/fiat market.
- Purchase the quoted crypto asset using fiat locked in Sendaza.
- Record the customer's crypto ownership in the Sendaza ledger.
- Reserve matching asset-network inventory from the SLE treasury.
- Withdraw the asset to a verified address on an enabled network.
- Sign and broadcast the withdrawal through Fireblocks MPC custody.
- Reconcile Sendaza, SLE, Fireblocks, and blockchain records.

OUT OF SCOPE Customer crypto deposits, crypto selling, swaps, staking, lending, automated treasury trading, and mainnet operation.

3. Ownership Boundary

Sendaza owns	SLE owns
Customer accounts, authentication, KYC, and transaction PIN	Executable crypto quotes and market configuration
Fiat funding and customer balances	Market-data conversion, spreads, purchase fees, and executable quotes
Display of SLE quote economics without recalculation	Treasury holdings, reservations, and sellable inventory
Available and locked balance accounting	Fireblocks MPC custody integration
Double-entry ledger and transaction history	External withdrawal submission and lifecycle
Customer-facing mobile and administrative interfaces	Blockchain confirmation and replacement tracking
Customer authorization and eligibility decisions	Treasury-to-liability reconciliation evidence

Non-negotiable boundary: SLE cannot write Sendaza customer balances, and Sendaza cannot access treasury signing keys.

4. Purchase Flow

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Customer selects an enabled crypto/fiat market and enters a fiat amount
→ Sendaza checks eligibility
→ Sendaza requests an executable quote from SLE
→ SLE evaluates the configured conversion route, spread, fees, limits, and sellable inventory
→ Customer confirms before quote expiry
→ Sendaza locks the exact fiat debit
→ SLE reserves the quoted crypto amount on its configured backing network
→ Sendaza posts balanced fiat and crypto ledger entries
→ Sendaza acknowledges settlement to SLE
→ Both systems mark the purchase complete
```

No blockchain transaction occurs during a purchase. The asset remains in the controlled treasury; the Sendaza ledger records customer ownership, and SLE reduces unallocated inventory by the same amount.

Pricing engine

SLE owns the complete, immutable quote calculation. Pricing routes are configuration-driven rather than hardcoded to ETH or NGN. Examples include $\text{BTC/USDT} \times \text{USDT/USD} \times \text{USD/NGN}$, $\text{SOL/USD} \times \text{USD/GHS}$,

or a reliable direct market. SLE stores every source rate, timestamp, conversion step, spread, fee, configuration version, and final customer amount. Sendaza displays the result and must not recalculate it.

5. Withdrawal Flow

Customer enters amount and Ethereum address

- SLE validates the destination and estimates fees
- Customer confirms the total debit
- Sendaza locks the withdrawal asset and applicable fees
- SLE evaluates limits, risk, treasury, and gas
- SLE submits one transfer instruction to Fireblocks
- Fireblocks policy approves automatically or requests review
- Fireblocks MPC signs and broadcasts
- SLE verifies the transaction through Alchemy
- Sendaza finalizes the ledger deduction after required confirmations

A submission timeout is not treated as a failure. Funds remain locked while SLE resolves the provider transfer using the original unique reference.

6. MPC Custody Model

Fireblocks creates and protects the treasury wallet using multi-party computation. SLE never receives or reconstructs the wallet's private key. SLE sends authenticated transfer instructions and receives provider transfer IDs and blockchain transaction hashes.

SLE stores	SLE never stores
Fireblocks vault/wallet identifier	Treasury private key
Public Ethereum address	Seed phrase or exportable signing material
Provider transfer identifier	Customer-facing custody credentials
Transaction hashes and status evidence	Secrets in source code, logs, or PostgreSQL

Fireblocks API authentication credentials are held in encrypted deployment secrets. These credentials allow SLE to request operations but are not the treasury wallet key.

Approval policy

- Low-risk withdrawals within approved limits may be processed automatically.
- High-value, unusual, or elevated-risk withdrawals require manual review or multiple approvals.
- Blocked destinations, insufficient treasury, invalid locks, or unavailable required screening stop the withdrawal.
- Fireblocks policies independently restrict source wallets, assets, networks, destinations, and amounts.

7. Sendaza Changes Required

1. **Asset precision:** support each configured fiat and crypto precision using atomic units or an approved exact representation.
2. **Asset model:** distinguish asset from network. For example, USDT on Ethereum, Tron, Solana, and Polygon are separate treasury and withdrawal routes.
3. **Ledger locking:** represent available-to-locked movements with balanced, immutable journals.
4. **Settlement protocol:** support durable lock, settle, release, and reconciliation states.
5. **SLE client module:** sign authenticated requests and attach idempotency and correlation identifiers.
6. **Inbound events:** verify SLE webhook signatures, timestamps, and replay identifiers before processing.
7. **Durable messaging:** write financial state and outbound acknowledgements into a transactional outbox.
8. **Operation records:** retain SLE order ID, client reference, lock reference, settlement reference, and transaction hashes.
9. **Liability reporting:** submit ledger-derived, cutoff-aligned liability snapshots for every supported asset.
10. **Provider separation:** remove direct crypto pricing, signing, and broadcasting from Sendaza once SLE is integrated.

Customer ownership may be displayed as USDT, but SLE tracks available inventory, token contracts, provider wallets, fees, and limits separately for every supported network. Inventory on one chain cannot

fund a withdrawal on another chain until a confirmed rebalancing operation has completed.

8. Integration Contract

Operation	Endpoint
Create purchase quote	POST /api/v1/quotes
Reserve quoted inventory	POST /api/v1/purchases
Acknowledge ledger settlement	POST /api/v1/purchases/:id/settlement
Retrieve purchase status	GET /api/v1/purchases/:id
Create withdrawal fee quote	POST /api/v1/withdrawal-fee-quotes
Submit approved withdrawal	POST /api/v1/withdrawals
Retrieve withdrawal status	GET /api/v1/withdrawals/:id
Report customer liabilities	POST /api/v1/reconciliation/liability-snapshots
Receive lifecycle events	Registered Sendaza webhook endpoint

All mutations require an idempotency key and correlation ID. Monetary values cross APIs as decimal strings. The MVP accepts only authenticated Sendaza credentials; external client onboarding is deferred.

9. Failure and Safety Rules

- A quote is not a purchase and does not reserve treasury inventory.
- A duplicate request cannot reserve or withdraw funds twice.
- An expired quote cannot be accepted.
- A provider timeout after submission does not prove failure.
- A broadcast transaction is not considered confirmed.
- Funds may be automatically released only after proven failure before broadcast.
- Uncertain results remain locked and enter reconciliation.
- Completed financial records remain immutable; corrections use compensating records.

10. Delivery Plan

Phase	Deliverable	Exit condition
1. Foundation	SLE service, PostgreSQL schema, client authentication, exact amounts, idempotency, audit log, and outbox	Concurrent and duplicate requests preserve financial invariants
2. Purchase	Configurable pricing routes, spreads, fees, treasury snapshots, reservations, and Sendaza settlement handshake	Sandbox purchase completes and reconciles without overselling
3. Withdrawal	Asset-network fee snapshots, Fireblocks adapter, provider webhooks, polling, and testnet finality	A testnet asset reaches an external address exactly once
4. Integration	Sendaza ledger changes, SLE client, lifecycle events, and liability snapshots	End-to-end purchase and withdrawal survive retries and restarts
5. Hardening	Reconciliation, security testing, incident exercises, and controlled pilot	No unexplained variance or unresolved critical/high finding

11. Decisions Required From Sendaza

- Approve the Sendaza/SLE ownership boundary.
- Approve the asset registry, precision rules, and ledger-account migration.
- Approve journal-backed locking, settlement, and reversal behavior.
- Decide purchase spread and explicit fee presentation.
- Decide withdrawal network-fee treatment and variance ownership.
- Approve automatic withdrawal limits and manual-review thresholds.
- Approve confirmation requirements by transaction value.
- Approve the liability snapshot definition and reporting cadence.
- Nominate engineering, treasury, compliance, and operations owners.

12. Acceptance Criteria

The sandbox integration is accepted when:

- A test customer buys a configured crypto asset using locked test fiat.
- Sendaza records exact, balanced, immutable ledger entries.
- SLE reserves sufficient asset-network inventory without overselling the treasury.
- The customer withdraws a supported testnet asset through Fireblocks MPC custody.
- Sendaza finalizes the deduction only after required confirmations.
- Duplicate, timeout, delayed-webhook, and restart tests do not duplicate or lose value.

- Sendaza, SLE, Fireblocks, and blockchain evidence reconcile at a common cutoff.

13. Approval Record

Area	Owner	Decision / date
Product and scope		
Engineering architecture		
Ledger and finance		
Custody and treasury		
Compliance and legal		
Security and operations		

Supporting design documents: PRD, Architecture, API Specification, Ledger Integration, Security and Operations, Technology Stack MVP, and Roadmap.